What does genetics have do with me? 9a

Has anyone ever told you that you have your father's nose or your grandmother's hair? Many of our features are inherited from our parents, and theirs from their parents. **Inheritance** or **heredity** is the passing of **characteristics** from parent to offspring – you. Characteristics include eye colour, hair colour, height, nose shape – and many more. We all inherit slightly different combinations of these features from our parents. Even brothers and sisters look different, because they inherit different characteristics from their parents.

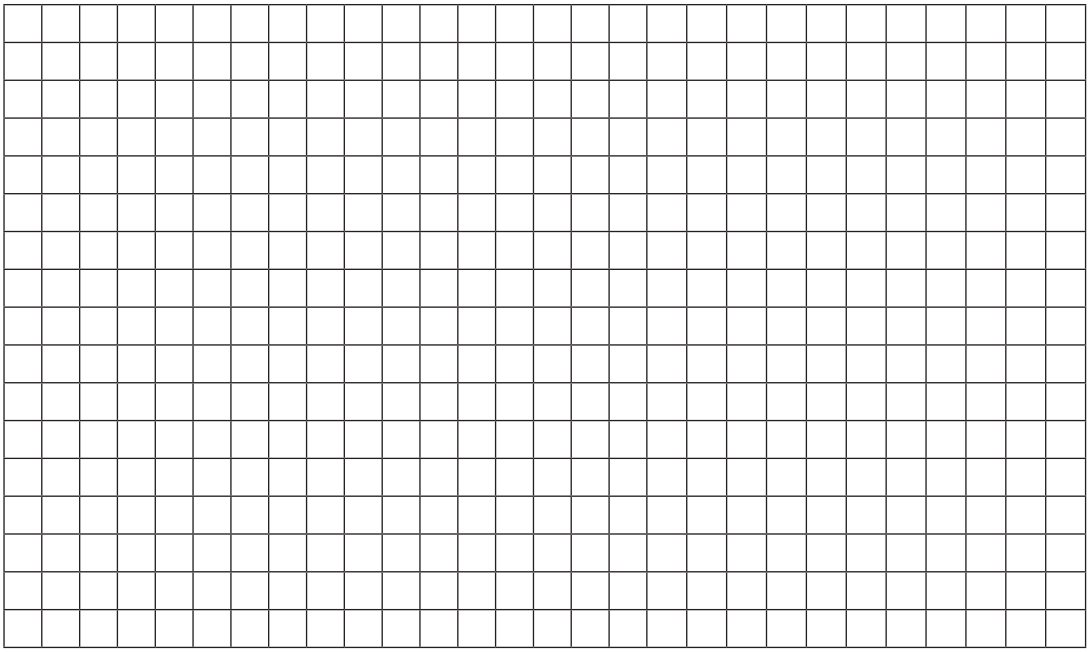
1 The table below shows data for characteristics collected from 30 year 9 students. Some of the 30 students did not respond for some characteristics. Add up each tally and calculate the percentage of students with that variation in characteristic. One example has been started for you. *Hint:* The percentages for each variation of a particular characteristic should add up to 100; e.g. 73% + 27% = 100%.

Characteristic	Variations in characteristic	Tally	Tally total	Total number of students	Percentage
Tongue rolling	*Yes, can*	𝍸 𝍸 𝍸 𝍸 ||	22	30	(22 ÷ 30) × 100 = 73%
	No, cannot	𝍸 |||		30	(8 ÷ 30) × 100 = 27%
Attached earlobes	*Yes, attached*	𝍸 𝍸 𝍸			
	No, unattached	𝍸 𝍸			
Receding hair on the sides of the forehead	*Yes, present*	𝍸 𝍸 𝍸			
	No, not present	𝍸 𝍸 𝍸			
Gap between front teeth	*Yes, gap*	𝍸			
	No, no gap	𝍸 𝍸 𝍸 𝍸			
Right thumb on top during hand folding	*Yes, right thumb on top*	𝍸 𝍸 𝍸 𝍸			
	No, left thumb on top	𝍸			
Hair on the second digits of your fingers	*Yes, present*	𝍸 𝍸 𝍸			
	No, not present	𝍸 𝍸			
Right dominant hand	*Yes, right*	𝍸 𝍸 𝍸 𝍸			
	No, left	𝍸			
Second toe longer than big toe	*Yes, longer*	𝍸 𝍸 𝍸			
	No, shorter	𝍸 𝍸 𝍸			
Straight hair	*Yes, straight*	𝍸 𝍸 𝍸 𝍸			
	No, not straight	𝍸 𝍸			

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What does genetics have do with me? 9b

2 Use the percentages you calculated on the previous page to construct a column graph of the results on the grid below.



3 Make two general statements about the graph.

4 Which of the more frequent variations do you possess?

Sometimes characteristics can be influenced by the environment. For example, consider two people with almost identical skin colour – one person likes to be outdoors and the other prefers to spend time indoors. These environments will affect their skin colour. Eating habits can impact on how tall people grow. Poor nutrition can inhibit growth, therefore a person with poor eating habits might not reach full height. Even identical twins can appear different if they are raised in different environments.

5 Which characteristics in the table on the previous page can be altered or influenced by the environment? Explain how.

6 Think about your own development. Do you think it has been influenced more by the environment or more by inheritance? Justify your answer using examples.

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